

EduGenie Learning Assistant – ER Diagram Documentation

Overview

The EduGenie Learning Assistant database supports AI-powered educational services including question answering, concept explanation, quiz generation, educational summarization, and personalized learning recommendations.

Entities

USER, USER_QUERY, AI_RESPONSE, QUIZ, SUMMARY, LEARNING_PATH

Relationships

USER → USER_QUERY (1:M)

USER_QUERY → AI_RESPONSE (1:1)

USER_QUERY → QUIZ (1:M)

USER_QUERY → SUMMARY (1:M)

USER_QUERY → LEARNING_PATH (1:M)

Normalization

The database follows normalization principles by separating users, educational requests, AI responses, quizzes, summaries, and learning recommendations into independent entities.

Use Case Coverage

Supports user management, educational question answering, concept explanations, quiz generation, summarization, learning recommendations, and AI response tracking.

Scalability

Compatible with PostgreSQL, MySQL, MongoDB, and Firebase. Future enhancements include student progress tracking, multilingual support, LMS integration, mobile applications, analytics dashboards, and adaptive learning systems.